

Adaptive Relational Trust System for Dynamic Network Integrity (ARTS-DNI)

Revised Research Concept and Technical Positioning

1. Executive Summary

ARTS-DNI proposes a transition from static credential-based trust toward continuous relational identity verification. Rather than treating trust as a point-in-time cryptographic event, the framework models trust as a dynamic trajectory evolving across time, topology, and behavioral state space. The system augments — not replaces — traditional cryptographic systems by introducing a continuous trust analytics layer capable of detecting topological anomalies, stealth MITM insertion, trust poisoning, and relational drift even when cryptographic primitives remain uncompromised.

2. Core Research Shift

Traditional cybersecurity systems evaluate identity through possession of valid credentials. ARTS-DNI introduces a different assumption: Trust = continuity of relational identity trajectory. In this model, identity is not a static credential but a statistically coherent evolution of: physical entropy signals, behavioral consistency, and graph-based interaction history.

3. Recommended Terminology Adjustments

To improve scientific rigor and reduce ambiguity, the following terminology adjustments are recommended:

- Replace “Device DNA” with “Physical Identity Anchor”
- Replace “offspring identity” with “relational session embedding”
- Replace biological metaphors with graph-theoretic terminology
- Position the framework as a trust analytics layer rather than a replacement for cryptography

4. System Architecture

The architecture consists of five primary layers:

1. Physical Entropy Layer - clock jitter - SRAM startup patterns - voltage fluctuation - RF timing noise
2. Cryptographic Trust Layer - ephemeral key exchange - hardware attestation - secure enclaves - PUF-derived signatures
3. Relational Trust Layer - dynamic trust graph $G(t)$ - edge lineage tracking - temporal trust continuity
4. AI Analytics Layer - anomaly scoring - topology mutation detection - entropy drift analysis - temporal graph embeddings
5. Human Oversight Layer - arbitration - policy validation - trust override decisions

5. Mathematical Positioning

The framework defines trust as a function of entropy stability, lineage continuity, and contextual consistency. $T_{AB}(t) = \sigma(\alpha * L_{AB}(t) - \beta * H_{AB}(t) + \gamma * C_{AB}(t))$ where: $L_{AB}(t)$ = relational continuity, $H_{AB}(t)$ = entropy instability, $C_{AB}(t)$ = behavioral consistency. This transforms trust evaluation from static verification into temporal state-space analysis.

6. Threat Model

The framework is designed to detect adversaries that preserve valid cryptographic states while altering network relational structure. Threat classes include: • stealth MITM insertion • topology poisoning • oscillatory trust attacks • Sybil-based graph manipulation • low-and-slow latency perturbation The adversary is assumed to possess protocol visibility but not the ability to perfectly replicate physical entropy characteristics.

7. Prototype Evaluation

Early proof-of-concept experiments using Isolation Forest anomaly detection demonstrated successful identification of low-amplitude latency manipulation attacks within noisy environments. Key observations: • successful detection of low-and-slow perturbations • adaptive thresholding reduced false positives • EWMA smoothing improved temporal stability • covariance regularization improved numerical robustness Current implementations remain statistical prototypes and do not yet constitute a full relational graph intelligence engine.

8. Current Limitations

Several critical limitations remain: • no formal security proof • no convergence or stability theorem • graph reasoning remains limited in current prototype • synthetic datasets dominate evaluation • relational embeddings are not yet implemented using Temporal GNNs The framework should therefore be considered an exploratory cyber-physical trust architecture rather than a finalized security protocol.

9. Research Roadmap

Recommended next-stage development: • Temporal Graph Neural Networks (TGN) • decentralized Local Trust Communities (LTC) • adversarial simulation environments • formal trust stability analysis • graph poisoning resilience evaluation • real-world mesh network deployment Priority should be placed on formalizing: detection guarantees, adversarial bounds, and trust continuity metrics.

10. Strategic Positioning

The strongest contribution of ARTS-DNI is not replacing cryptography, but augmenting existing trust systems with: • temporal relational integrity analysis • topology-aware trust modeling • continuity-based anomaly detection The concept aligns strongly with: • adaptive zero-trust systems • cyber-physical identity frameworks • distributed autonomous networks • resilient edge infrastructure • autonomous swarm coordination

11. Final Assessment

Category	Assessment
Vision & Novelty	Strong exploratory research concept
Scientific Rigor	Needs formal proofs and comparative evaluation
Prototype Maturity	Early-stage statistical proof-of-concept
Research Potential	High potential for DARPA-style exploratory programs
Best Positioning	Trust analytics augmentation layer